

Business & Functional Requirements Specification

Knowledge Management System (KMS)

*Document & File Storage, Search, and Retrieval Platform for
Internal Use*

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Table of Contents

Table of Contents	2
1. Introduction	3
1.1 Purpose	3
1.2 Background	3
1.3 Objectives	3
2. Scope	3
3. Stakeholders & User Roles	4
4. Business requirements	4
5. Functional Requirements	5
5.1 Document & File Management	5
5.2 Search & Retrieval	7
5.3 Access Control & Security	7
5.4 Collaboration & Workflow	8
5.5 Administration & Reporting	8
6. Non-Functional Requirements	10
7. Integration Requirements	11
8. Roles & Permissions Matrix	11
9. Data Requirements	12
9.1 Required Metadata Fields	12

1. Introduction

1.1 Purpose

This document defines the business, functional, and non-functional requirements for an internal Knowledge Management System (KMS). The primary purpose of the system is to provide employees with a single, secure, and searchable repository for organizational documents and files, replacing fragmented storage across shared drives, email attachments, and individual desktops.

1.2 Background

Organizational knowledge is currently distributed across disconnected file shares, personal drives, email threads, and collaboration tools. This fragmentation results in duplicated effort, version confusion, lost institutional knowledge when staff leave, and significant time spent searching for existing information. A centralized KMS addresses these problems by providing a governed, searchable, and auditable home for internal documents and files.

1.3 Objectives

- Provide a single source of truth for internal documents and files.
- Enable fast, accurate full-text and metadata-based search across all stored content.
- Reduce time spent locating information and reduce duplicate
- Enforce consistent access control, versioning, and retention practices.
- Preserve institutional knowledge independent of individual staff turnover.
- Provide administrators with visibility into usage, storage, and compliance posture.

2. Scope

- Internal, employee-facing document and file storage, organization, and search.
- Upload, versioning, tagging, categorization, and lifecycle management of files.
- Role-based access control and permission management at folder and document level.
- Full-text search, filters, and metadata-driven discovery.
- Basic collaboration features: comments, sharing links, check-in/check-out.
- Audit logging, reporting, and administrative controls.

3. Stakeholders & User Roles

The following roles interact with the system. Detailed permission mappings are defined in Section 8.

Role	Description
System Administrator	Manages system configuration, storage, integrations, and permissions.
Content Owner / Manager	Owns a department or folder area; manages structure, permissions, and retention within it.
Contributor	Uploads, edits, and organizes documents within areas they have write access to.
Viewer	Searches for and reads/downloads documents; no upload or edit rights.
Compliance / Records Officer	Defines retention policies, runs audits, and manages legal holds.
IT Security	Monitors access logs, manages authentication policy, and investigates incidents.

4. Business requirements

High-level business needs the system must satisfy, independent of implementation detail.

ID	Requirement	Description	Priority
BR-01	Centralized Repository	Provide one authoritative platform for storing internal documents and files, reducing reliance on shared drives and email.	Must
BR-02	Findability	Employees must be able to locate relevant documents in seconds using search rather than folder navigation alone.	Must

ID	Requirement	Description	Priority
BR-03	Governance	Support consistent classification, access control, and retention aligned with the organization policy.	Must
BR-04	Continuity	Preserve organizational knowledge when employees change roles or leave the organization	Must
BR-05	Adoption	System must be usable enough that employees prefer it over ad hoc storage (shared drives, local desktops).	Should
BR-06	Cost Efficiency	Reduce storage sprawl and duplicate copies of the same document across systems.	Should
BR-07	Auditability	Provide a defensible audit trail of who accessed, modified, or deleted content, for compliance and investigations.	Must

5. Functional Requirements

5.1 Document & File Management

ID	Requirement	Description	Priority
FR-01	File Upload	Users can upload individual files or bulk-upload folders via browser drag-and-drop, desktop sync client, or API.	Must

ID	Requirement	Description	Priority
FR-02	Supported File Types	System supports common office formats (docx, xlsx, pptx), PDF, images, CAD, audio/video, and compressed archives.	Must
FR-03	Folder & Taxonomy Structure	Support hierarchical folders plus non-hierarchical tags/categories/metadata for flexible organization.	Must
FR-04	Versioning	Automatic version history on every save/upload, with ability to view, compare, restore, and annotate prior versions.	Must
FR-05	Check-in / Check-out	Users can lock a document for exclusive editing to prevent conflicting concurrent changes.	Should
FR-06	Metadata & Tagging	Custom and required metadata fields (department, document type, confidentiality level, owner, effective date) attachable to each file.	Must
FR-07	Bulk Operations	Support bulk move, tag, delete, and permission changes across multiple selected items.	Should
FR-08	Recycle Bin / Recovery	Deleted items are recoverable within a configurable retention window before permanent purge.	Must
FR-09	File Preview	In-browser preview of common file types without requiring download or native application.	Should
FR-10	OCR & Text Extraction	Scanned documents and images with embedded text are OCR-processed to make content full-text searchable.	Should

5.2 Search & Retrieval

ID	Requirement	Description	Priority
FR-11	Full-Text Search	Search indexes both file names/metadata and extracted document content across all supported formats.	Must
FR-12	Filters & Facets	Search results can be filtered by file type, owner, department, date range, tags, and confidentiality level.	Must
FR-13	Advanced Query Syntax	Support boolean operators, exact phrase matching, and wildcard search for power users.	Could
FR-14	Relevance Ranking	Results are ranked by relevance, incorporating recency and popularity signals, with manual re-sort options.	Should
FR-15	Saved Searches & Alerts	Users can save frequent searches and opt in to be notified when new matching content is added.	Could
FR-16	Permission-Aware Results	Search results only ever display content the requesting user is authorized to see.	Must

5.3 Access Control & Security

ID	Requirement	Description	Priority
FR-17	Role-Based Access Control	Permissions (view, edit, upload, delete, manage) assignable at folder and individual document level by role or group.	Must
FR-18	Single Sign-On	Authentication via the corporate identity provider (SAML/OIDC), including MFA where enforced organization-wide.	Must
FR-19	Confidentiality Labels	Documents can be classified (e.g., Public, Internal, Confidential, Restricted) with access enforced by label.	Must

ID	Requirement	Description	Priority
FR-20	Secure Sharing Links	Time-limited, revocable, optionally password-protected links for sharing with users outside standard access groups.	Should
FR-21	Encryption	Data encrypted at rest and in transit using current industry-standard protocols.	Must
FR-22	Audit Logging	All view, edit, download, share, permission-change, and delete events are logged with user, timestamp, and IP.	Must

5.4 Collaboration & Workflow

ID	Requirement	Description	Priority
FR-23	Comments & Annotations	Users can comment on documents and reply in threads without altering the underlying file.	Should
FR-24	Native App Integration	Documents open directly in their native productivity application (e.g., Word, Excel) for editing, then sync back automatically.	Must
FR-25	Approval Workflow	Optional simple approval routing (e.g., draft → review → published) for designated document types.	Could
FR-26	Notifications	Users can subscribe to folders/documents and receive notifications on new versions, comments, or shares.	Should

5.5 Administration & Reporting

ID	Requirement	Description	Priority
FR-27	Admin Console	Central console for managing users, groups, storage quotas, retention policies, and system configuration.	Must

ID	Requirement	Description	Priority
FR-28	Retention & Disposition	Configurable retention schedules by document type/category, with automated archival or deletion workflows.	Must
FR-29	Legal Hold	Ability to place a hold on specific content to suspend normal retention/deletion during litigation or investigation.	Should
FR-30	Usage & Storage Reporting	Dashboards showing storage consumption, active users, top searches, and content growth trends.	Should
FR-31	Orphaned/Stale Content Reports	Reports flagging documents with no owner, no recent access, or expired review dates for cleanup.	Could

6. Non-Functional Requirements

ID	Requirement	Description	Priority
NFR-01	Performance	Search results returned within 2 seconds for 95% of queries under normal load.	Must
NFR-02	Scalability	System must support growth to at least 3x current document volume and concurrent user count without redesign.	Must
NFR-03	Availability	99.5% uptime during business hours, excluding scheduled maintenance windows communicated in advance.	Must
NFR-04	Usability	New employees should be able to find and upload a document within their first session with minimal training.	Should
NFR-05	Accessibility	Web interface complies with WCAG 2.1 AA standards.	Should
NFR-06	Data Durability	No data loss on stored content; automated backups with a defined recovery point objective (RPO) and recovery time objective (RTO).	Must
NFR-07	Compliance	System configuration supports applicable data protection and records-retention regulations relevant to the organization's jurisdiction(s).	Must
NFR-08	Mobile Access	Core browse, search, and view functions usable from a mobile browser or dedicated app.	Could
NFR-09	Maintainability	System configuration (permissions, taxonomy, retention rules) manageable by administrators without vendor engineering support.	Should

ID	Requirement	Description	Priority
NFR-10	Interoperability	Exposes an API for integration with other internal systems and future automation.	Should

7. Integration Requirements

System	Integration Purpose	Priority
Corporate Identity Provider (SSO/MFA)	Authentication and single sign-on for all users	Must
Productivity Suite (e.g., Microsoft 365 / Google Workspace)	Native open/edit/save-back for office documents	Must
Email System	Send notifications, alerts, and share links	Must
HRIS	Sync organizational structure, groups, and offboarding for access de-provisioning	Should
Chat/Collaboration Platform (e.g., Slack/Teams)	Share links and receive notifications in existing team channels	Could
SIEM / Security Monitoring	Forward audit logs for centralized security monitoring	Should

8. Roles & Permissions Matrix

Action	Viewer	Contributor	Content Owner	Administrator
View / Search / Download	Yes	Yes	Yes	Yes
Upload / Edit	No	Yes	Yes	Yes
Delete	No	Own uploads only	Yes (within area)	Yes

Action	Viewer	Contributor	Content Owner	Administrator
Manage Permissions	No	No	Yes (within area)	Yes
Configure Retention/Legal Hold	No	No	Request only	Yes
System Configuration	No	No	No	Yes

9. Data Requirements

9.1 Required Metadata Fields

- Document title and unique identifier
- Owner / responsible department
- Document type / category (e.g., Policy, Contract, Report, Template)
- Confidentiality classification
- Creation date, last modified date, last modified by
- Version number
- Retention category / review date